

References for the KM Complex Filter Toolbox

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This document lists books, papers, reports and manuals that were used in developing this toolbox but cannot be included in the repository because of copyright. Each entry gives the full reference and a short note on how it relates to the toolbox. IEEE papers can be obtained from IEEE Xplore using the reference or the DOI given. Closely related work that is freely available online is listed, with links, in its own section. Documents that can be shared are in the repository's `doc` folder and are listed in the last section.

The bibliographic details were compiled from the documents themselves. Details that could not be confirmed that way are marked *to be confirmed*. The list will be extended as the work continues; it is generated from `doc/References.yaml` by `tools/make_references.py`.

1 Complex Signals and Complex Filters

- [1] K. W. Martin, “Complex signal processing is not complex,” *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 51, no. 9, pp. 1823–1836, Sept. 2004.
Relevance: Tutorial on complex signals and complex transfer functions in wireless systems, including complex filters, image-reject mixers and low-IF receivers. The background for everything in the toolbox.
- [2] W. M. Snelgrove, “Intermediate-function synthesis,” Ph.D. dissertation, Dept. Elect. Eng., Univ. of Toronto, Toronto, ON, Canada, 1982.
Relevance: A vector-space approach to the design of active filters, with algorithms for computer-aided design. Background from the group that introduced complex filters (see the Snelgrove et al. report in the last section).
- [3] C. Muto, “A new extended frequency transformation for complex analog filter design,” *IEICE Trans. Fundamentals*, vol. E83-A, no. 6, pp. 934–940, June 2000.
Relevance: A frequency transformation that maps arbitrary passband and stopband edges of a complex analog filter onto a normalized lowpass filter. An alternative to the transformed-variable approach used in the toolbox.
- [4] L. Yu and W. M. Snelgrove, “A novel adaptive mismatch cancellation system for quadrature IF radio receivers,” *IEEE Trans. Circuits Syst. II*, vol. 46, no. 6, pp. 789–801, June 1999.
Relevance: I/Q gain and phase mismatch in quadrature IF receivers, and its adaptive cancellation. Application context for complex filters.

2 Approximation of Complex IIR Filters, Transformed Variables and Group Delay

- [5] K. W. Martin, “Approximation of complex IIR bandpass filters without arithmetic symmetry,” *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 52, no. 4, pp. 794–803, Apr. 2005. DOI: [10.1109/TCSI.2005.844363](https://doi.org/10.1109/TCSI.2005.844363).
Relevance: The approximation algorithms on which the toolbox’s continuous-time design routines are based: equi-ripple and monotonic passbands, arbitrary stopbands, and loss-pole placement.
- [6] H. J. Orchard and G. C. Temes, “Filter design using transformed variables,” *IEEE Trans. Circuit Theory*, vol. CT-15, no. 4, pp. 385–408, Dec. 1968.
Relevance: The classical paper on filter approximation in transformed variables. The toolbox’s z -to- y and band-centred z -to- x transforms follow this approach.
- [7] K. Martin, “Direct approximation of complex IIR filters,” manuscript, Granite SemiCom Inc., Toronto, ON, Canada, 2016.
Relevance: The author’s manuscript on approximating complex IIR filters directly, for both continuous-time and sampled filters.
- [8] K. Martin, “Equiripple group-delay IIR digital filters,” manuscript, Granite SemiCom Inc., Toronto, ON, Canada, 2019.
Relevance: The author’s manuscript on equi-ripple group-delay digital filters obtained from continuous-time prototypes with a bilinear transform. Related to the toolbox’s `dig_equiGd` and `dig_linPh` designs.
- [9] M. Alijan, “DSP compensation for distortion in RF filters,” M.Sc. thesis, Dept. Elect. Comput. Eng., Univ. of Saskatchewan, Saskatoon, SK, Canada, Apr. 2010.
Relevance: Compensates the asymmetric (hence complex-baseband) amplitude and group-delay distortion of the RF bandpass filter in a CATV QAM modulator: a complex FIR filter for the amplitude, and a low-order complex all-pass IIR filter for the group delay, whose poles are optimized by a quasi-Newton method with the radii limited to 0.99. Reports that the group-delay optimization is much less effective than the amplitude optimization and very sensitive to the pole positions. The closest earlier complex-filter group-delay equalizer; no code or coefficients are given.
- [10] D. Živaljević, N. Stamenković, and J. Živanić, “Sharp cutoff filters with monotonic passband response,” in *Proc. IEEE Conf.*, 2014 (conference name to be confirmed; ISBN 978-1-4799-5817-7).
Relevance: Filters with monotonic passbands and sharp cutoff. Related to the toolbox’s monotonic-passband approximation option.

3 Filter Realization and Design Tools

- [11] K. Martin, “Bilinear- z digital filters based on SFGs,” manuscript, Granite SemiCom Inc., Toronto, ON, Canada, 2018.
Relevance: Applying the bilinear- z transform to integrator-based signal-flow-graph filters, and how the resulting delay-free loops can be handled.

- [12] K. Martin, “Complex ladder filter realizations,” manuscript, Granite SemiCom Inc., Toronto, ON, Canada, 2018.
Relevance: Extends passive ladder synthesis to transfer functions with non-conjugate loss poles, that is, to complex filters.
- [13] C. Ouslis, M. Snelgrove, and A. S. Sedra, “A filter designer’s filter design aid: filterX,” 1991 (venue to be confirmed).
Relevance: The filterX filter-design language, with minimax optimization of filter responses. An early example of an integrated filter design tool.
- [14] B. M. Silveira, C. Ouslis, and A. S. Sedra, “Passive ladder synthesis in filterX,” in *Proc. IEEE Int. Symp. Circuits Syst. (ISCAS)*, 1993, pp. 1431–1434.
Relevance: Ladder-synthesis primitives in filterX, including a ladder derived from a bandpass transfer function.

4 Filter Banks and Transmultiplexers

- [15] M. Padmanabhan and K. Martin, “Resonator-based filter-banks for frequency-domain applications,” *IEEE Trans. Circuits Syst.*, vol. 38, no. 10, pp. 1145–1159, Oct. 1991.
Relevance: Filter banks based on a digital simulation of a singly terminated ladder, with $O(N)$ complexity. Background for the toolbox’s filter bank examples.
- [16] K. Martin and M. Padmanabhan, “Resonator-in-a-loop filter banks based on a Lerner grouping of outputs,” in *Proc. IEEE Int. Conf. Acoust., Speech, Signal Process. (ICASSP)*, 1992, vol. IV, pp. IV-329–IV-332.
Relevance: Resonator-in-a-loop filter banks with alternating-sign (Lerner) grouping of the outputs, giving much lower side lobes.
- [17] M. Padmanabhan and K. Martin, “Some further results on modulated/extended lapped transforms,” in *Proc. IEEE Int. Conf. Acoust., Speech, Signal Process. (ICASSP)*, 1992, vol. IV, pp. IV-265–IV-268.
Relevance: Perfect-reconstruction conditions for modulated and extended lapped transforms, and an efficient filter structure for them.
- [18] M. Padmanabhan and K. Martin, “Filter banks for time-recursive implementation of transforms,” *IEEE Trans. Circuits Syst. II*, vol. 40, no. 1, pp. 41–50, Jan. 1993.
Relevance: A general filter-bank structure that implements an arbitrary transform recursively in time, with low complexity for the DFT and DCT.
- [19] K. W. Martin, “Small side-lobe filter design for multitone data-communication applications,” *IEEE Trans. Circuits Syst. II*, vol. 45, no. 8, pp. 1155–1161, Aug. 1998.
Relevance: Channel-bank filters built from weighted sums of neighbouring inverse Fourier transform filters, with much better stopbands.
- [20] S. Mirabbasi and K. Martin, “Design of prototype filter for near-perfect-reconstruction overlapped complex-modulated transmultiplexers,” in *Proc. IEEE Int. Symp. Circuits Syst. (ISCAS)*, 2002, vol. 1. DOI: [10.1109/ISCAS.2002.1009967](https://doi.org/10.1109/ISCAS.2002.1009967).
Relevance: Prototype-filter design for overlapped complex-modulated transmultiplexers.

- [21] S. Mirabbasi and K. Martin, “Overlapped complex-modulated transmultiplexer filters with simplified design and superior stopbands,” *IEEE Trans. Circuits Syst. II*, vol. 50, no. 8, pp. 456–469, Aug. 2003. DOI: [10.1109/TCSII.2003.813592](https://doi.org/10.1109/TCSII.2003.813592).
Relevance: The journal version: a unified prototype-filter design for all overlap factors, with high stopband attenuation and fast side-lobe fall-off.
- [22] T. Saramäki and R. Bregović, “Multirate systems and filter banks,” in *Multirate Systems: Design and Applications*, G. Jovanovic-Dolecek, Ed. Hershey, PA: Idea Group Publishing, 2002, ch. 2.
Relevance: A survey of multirate systems and filter banks.

5 Numerical Methods for Polynomial Roots

- [23] W. H. Press, S. A. Teukolsky, W. T. Vetterling, and B. P. Flannery, *Numerical Recipes in C*, Cambridge, U.K.: Cambridge Univ. Press, Sec. 9.5, “Roots of polynomials” (edition to be confirmed).
Relevance: Polynomial root finding and the ill-conditioning of high-order polynomials. Accurate root finding matters throughout the toolbox, which works with poles and zeros rather than polynomial coefficients.
- [24] M. Lang and B.-C. Frenzel, “A new and efficient program for finding all polynomial roots,” Tech. Rep. 9308, Dept. Elect. Comput. Eng., Rice Univ., Houston, TX, USA, Apr. 1994.
Relevance: A fast and accurate program for finding all the roots of a polynomial.
- [25] Textbook excerpt, ch. 2, “Solution of nonlinear equations $f(x) = 0$,” section on Muller’s method, pp. 92–96 (source book to be confirmed).
Relevance: Muller’s method, used in the toolbox to refine polynomial roots when solving Feldtkeller’s equation.
- [26] Y.-B. Jia, “Roots of polynomials,” Com S 477/577 course notes, Iowa State Univ., Ames, IA, USA, 2018.
Relevance: Course notes on polynomial root finding.
- [27] E. Pardyjak, “Roots of polynomials,” lecture 5 slides, 2006 (institution to be confirmed).
Relevance: Lecture slides on polynomial root finding.
- [28] M. Galassi *et al.*, *GNU Scientific Library Reference Manual*, release 2.5, 2018. Available: <https://www.gnu.org/software/gsl/>
Relevance: Reference manual for the GNU Scientific Library, including its polynomial root finders. Freely available from the URL.

6 Circuit and System Analysis

- [29] W. J. McCalla, *Fundamentals of Computer-Aided Circuit Simulation*. Boston, MA, USA: Kluwer Academic Publishers, 1988, ch. 7, “Pole-zero evaluation.”
Relevance: Computing the poles and zeros of circuit transfer functions, and the numerical problems involved.

- [30] *HSPICE User's Manual*, ch. 24, “Performing pole/zero analysis,” 1998 (publisher to be confirmed).
Relevance: Pole/zero analysis in a commercial circuit simulator.
- [31] K. J. Åström and R. M. Murray, *Feedback Systems*, draft, ch. 6, “Transfer functions,” Nov. 2004.
Relevance: Transfer functions and their relation to other system descriptions.

7 Related Work Available Online

- [32] J. Munir, A. Mezghani, I. Slim, and J. A. Nossek, “Chromatic dispersion compensation using complex-valued all-pass filter,” in *Proc. 22nd Eur. Signal Process. Conf. (EUSIPCO)*, Lisbon, Portugal, 2014, pp. 396–400. Available: <https://www.eurasip.org/Proceedings/Eusipco/Eusipco2014/HTML/papers/1569924013.pdf>
Relevance: Equalizes a complex-baseband all-pass channel (optical-fibre chromatic dispersion) with a cascade of first-order complex all-pass sections, optimized in pole radius and angle. The section form and its group delay are the same as in the toolbox’s cluster equalizers, and the paper is a clear, recent introduction to complex all-pass equalization. A filter-bank extension is at arXiv:1403.1732.
- [33] J. S. Abel and J. O. Smith, “Robust design of very high-order allpass dispersion filters,” in *Proc. Int. Conf. Digital Audio Effects (DAFx-06)*, Montreal, QC, Canada, 2006, pp. 13–18. Available: https://www.dafx.de/paper-archive/2006/papers/p_013.pdf
Relevance: Fits very high-order all-pass cascades to a target group delay by dividing the band into regions of equal delay area, which also fixes the order. Related to the delay-area estimate of the number of all-pass sections used in the toolbox (`estAllPassOrder`).
- [34] M. F. Quélhas, A. Petraglia, and M. R. Petraglia, “Efficient group delay equalization of discrete-time IIR filters,” in *Proc. 12th Eur. Signal Process. Conf. (EUSIPCO)*, Vienna, Austria, 2004. Available: <https://www.eurasip.org/Proceedings/Eusipco/Eusipco2004/defevent/papers/cr1483.pdf>
Relevance: Pole-allocation estimates and computer-aided refinement for real-coefficient all-pass group-delay equalizers of elliptic filters.

8 Documents Included in the Repository

- [35] W. M. Snelgrove, A. S. Sedra, G. R. Lang, and P. O. Brackett, “Complex analog filters,” unpublished, 1981.
In the repository: `doc/snelgrove_cmplx.pdf`. The report that introduced the y^2 transformed variable for complex analog filters. Shared with the author’s permission.